



HINDUSTAN COPPER LIMITED
(A Govt. of India Enterprise)

Expression of Interest (Eoi) for

**RECOVERY OF MINERALS & MATERIAL FROM COPPER ORE TAILS AT
KHETRI COPPER COMPLEX, KHETRINAGAR, RAJASTHAN**

10 August 2013

Reference No: HCL/COT/EOI/2013
Last date of submission: 20.9.2013

Tamra Bhavan, 1, Ashutosh Chowdhury Avenue, Kolkata - 700 019



Expression of Interest For Recovery of Minerals & Material From Copper Ore Tails at Khetri Copper Complex, Khetrinagar, Rajasthan

1.0 General

1.1 Hindustan Copper Limited (HCL), a public sector undertaking under the administrative control of the Ministry of Mines, was incorporated on 9th November 1967. It has the distinction of being the Nation's only vertically integrated copper producing company as it manufactures copper right from the stage of mining to beneficiation, smelting, refining and casting of refined copper metal into downstream saleable products. HCL has also the distinction of owning all copper mining leases in the country.

1.2 The company has its corporate and registered office in Kolkata. HCL has four operating units viz. Khetri Copper Complex (Rajasthan), Malanjkhand Copper Project (Madhya Pradesh), Indian Copper Complex, Ghatsila (Jharkhand) and Taloja Copper Project (Maharashtra). While Khetri and Ghatsila are fully integrated units (mining, ore beneficiation, smelting and refining), Malanjkhand has mining and ore beneficiation and Taloja has wire rod manufacturing facility only. The marketing/sales setup is headquartered at Kolkata with four regional sales offices in Delhi, Mumbai, Kolkata & Bangalore and sales godowns are spread across the country. The company has posted a net profit after tax of Rs. 355.64 crore during 2012-13.

1.3 The company manufactures and markets copper cathodes, copper continuous cast copper wire rod and by-products such as anode slime (containing gold, silver & selenium), copper sulphate, slime and sulphuric acid.

Further details about the company and its operations can be obtained from its website www.hindustancopper.com.



2.0 Project Background:

2.1 Khetri Cooper Complex (KCC), Khetrinagar, is located in the District Jhunjhunu, Rajasthan. The complex currently produces 10.0 lakh tonne of copper ore from its mines having average copper content of grade 0.9%. The ore is subjected to ore beneficiation process (involving crushing, grinding, floatation, thickening and filtration) to produce Copper Concentrate, having around 17% copper content. In the beneficiation process, only 5 to 7% of ore quantity reports to concentrate and balance 93 to 95% of the ore reports to a tailing dam. Over a period of time, around 45.0 million tonne of tailings in the form of dense slurry has accumulated at KCC tailing pond. The complex is implementing mine expansion projects to increase its current mine capacity from 1.0 million tonne to 3.1 million tonne.

2.2 Copper ore tails (COT) is in powdered form (50 to 60% minus 200 mesh size). Its specific gravity is about 3.14 and contains 90 to 95% insoluble (mainly silica and iron) with very small quantity of other metals. Thus, COT is rich in metals, sand minerals and has the potential to be used in agriculture as a cheaper source of micronutrients, either in its present form or after suitable processing/refining. Micronutrients play a vital role in growth and developments of plants & are considered to be essential for increasing crop yields.

2.3 Chemical analysis of COT (on dry basis) as reported by Non-Ferrous Materials Technology Development Centre, Hyderabad along with ore mineralogy and flow sheet of KCC concentrator plant is given at enclosure -1.

2.4 HCL is interested in recovering the locked up minerals and materials in the above tails through a suitable process and technology.



3.0 Expression of Interest

3.1 Expression of Interest (EOI) is invited from the technically competent and reputed Metallurgical/Mineral Processing & Technology Development organizations and Engineering solution Suppliers to investigate, study, propose and execute a process route to recover entrapped Metal values in Tailings, either in the form of Minerals, Metals and/or a High Grade Bulk Concentrate (amenable to conventional Metallurgical Extraction processes) in an integrated, eco-friendly and cost effective manner. The expertise in solid-liquid separation technologies and equipment are also desired.

3.2 HCL is interested in selecting interested technology capable organization for the following:

- i) To provide Techno-Economically feasible process route and Basic Engineering for extraction of minerals from the tailings (from the material already dumped and online treatment of tails being produced).
- ii) Detailed Engineering and Construction of the Plant as per the Process Route selected.
- iii) Possible Business models for operation of the plant (including disposal of effluent) and marketing of the products extracted.

4.0 The interested parties are requested to visit site at Khetri Copper Complex, Khetrinagar, to familiarize themselves with the site conditions and contact Shri A K Ghosh, General Manager, Khetri Copper Complex, (Contact No. 01593 220001) for any assistance. Further clarifications, if any, may be sought with in regards to scope, technical feasibility etc. from the undersigned.

5.0 Interested organizations either as a single entity or in consortium shall submit their sealed EOI latest by 14:30 hours (IST) on or before 20th September, 2013, addressed to Mr. Vivek Gupta, Senior Manager, Hindustan



Copper Limited, “Tamra Bhavan” 1, Ashutosh Chowdhury Avenue, Kolkata 700 019 with following details:

- i) Brief description of their Organization, stating type of business, nature of company, location and details of management.
- ii) Experience details such as list of similar nature of jobs (Technology Development, EPC Projects etc.) carried out in last 5 years with client reference.
- iii) Annual reports/ Audited accounts for the last 3 years.
- iv) Write up on the Technical and Business proposals.

6.0 At any time prior to deadline for submission of proposals, HCL may for any reason, modify the Eol document. All such amendment shall become part of the Eol and same will be notified on Company’s website. The respondents are required to have a watch on Company’s website for any such amendment.

7.0 The Eol may be submitted by courier/speed post, alternatively, the Eol document may also be deposited by the respondent in the Tender Box located in 3rd Floor of HCL Corporate office by the representative of the respondent within the due date and time.

Vivek Gupta
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e-mail: vivek_g@hindustancopper.com
hindcopper@hotmail.com



Enclosure -1

Phones: 2434 1332 (5 Lines)
Direct : 2434 2300, 2434 2567

Fax : 0091-40-2434 0592
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NFTDC

NONFERROUS MATERIALS TECHNOLOGY DEVELOPMENT CENTRE

(A Registered Autonomous R&D Institution)
P.O. KANCHANBAGH, HYDERABAD – 500 058, INDIA

30th Aug.2010

CHEMICAL ANALYSIS REPORT

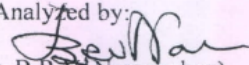
The chemical constituents of Copper Ore Tailing are given below:

S.No	Parameter / Element	KCC* – Copper Ore Tailings
1	Sp.Gravity	3.14
2	Bulk Density, g/ml	1.34
3	Cu	0.13 wt%
4	Al ₂ O ₃	4.53 wt%
5	Fe ₂ O ₃	16.96 wt%
6	CaO	0.70 wt%
7	MgO	1.65 wt%
8	K ₂ O	0.70 wt%
9	Na ₂ O	0.15 wt%
10	TiO ₂	0.13 wt%
11	Zn	36 PPM
12	Mn	890 PPM
13	Pb	17 PPM
14	As	81 PPM
15	Au	0.18 PPM
16	Ag	5.9 PPM
17	Mo	9 PPM
18	Sb	8 PPM
19	Ni	29 PPM
20	Co	40 PPM
21	Re	16 PPM
22	Se	0.9 PPM
23	Cd	1 PPM
24	Sn	70 PPM
25	Hg	23 PPM
26	S	1.31wt%
27	SiO ₂	73.54 wt%
28	Total Insolubles (aqu)	99.08wt %

*Analytical results are on DRY BASIS.

Trace metal analysis using ICP-OES (Teledyne Leeman)

Analyzed by:


(Dr. B.R.V. Narasimhan)
Sr.Scientist (Projects)
For NFTDC

Note:

1. The samples are drawn and submitted by the Client.
2. The Centre can't be attached legally and is not responsible for any dispute or loss arising out of the usage of this report



Mineralogy of the ore at KCC:

Mineral	Approximate %
Chalcopyrite	~3
Pyrite & Pyrrhotite	10-12
Magnetite	3-4
Quartz	45-50
Mica (Muscovite, Biotite & Chlorite)	25-30
Amphibole	5-7
Garnet	1-2
Feldspar	Traces
Calcite	Traces

[Flow Sheet of KCC Concentrator Plant](#)

